

How Russian speakers express evolution in Pokémon names: An experimental study*

Abstract

Sound symbolism, systematic and iconic relationships between sounds and meanings, is now a topic that is very actively explored by linguists, psychologists and cognitive scientists. As a new research strategy to study the nature of sound symbolic connections across different languages, a number of scholars have started using Pokémon names, a research paradigm that is now dubbed “Pokémonastics.” The previous Pokémonastic studies have experimentally explored how the evolution status is symbolically expressed by speakers of English, Japanese and Brazilian Portuguese. Building on these studies, the current experiment examined the sound symbolic knowledge of Russian speakers, and found that they are more likely to associate large, post-evolution Pokémon characters with names containing voiced obstruents than with names containing voiceless obstruents, and they are also more likely to associate post-evolution characters with names containing [a] than with names containing [i]. The experiment also revealed that Russian speakers are less likely to associate post-evolution characters with names having labial consonants than with names having coronal or dorsal consonants. Overall, the current results show that Russian speakers have knowledge of sound symbolic associations that are similar to those of English, Japanese and Brazilian Portuguese speakers, suggesting that some sound symbolic patterns hold robustly across multiple languages.

1 Introduction

Modern thinking about human languages often took it for granted that the relationships between sounds and meanings are at least in principle arbitrary (Saussure 1916/1972). However, there is a rapidly growing body of studies showing that there can be systematic correspondences between sounds and meanings (e.g. Dingemanse et al. 2015; Lockwood & Dingemanse 2015). When such

*The acknowledgments will go here.

6 sound-meaning associations are modulated via iconicity between sounds and meanings, those re-
7 lationships are referred to as “sound symbolism” (Hinton et al. 1994). Studying sound symbolic
8 connections is now considered to be an important topic for linguistic inquiry and cognitive science,
9 as such connections may guide language acquisition processes to non-trivial degrees (Imai & Kita
10 2014; Nygaard et al. 2009), and they may also bear on the question of how human languages may
11 have originated and evolved (Cuskley & Kirby 2013; Perlman & Lupyan 2018). The importance
12 of studying sound symbolic patterns for formal phonological theories has also been implicated by
13 various researchers (Alderete & Kochetov 2017; Jang 2021; Kawahara 2020; Kumagai 2019; Shih
14 2020). Finally, practical application of sound symbolism for areas of research beyond linguistics
15 and cognitive science, such as marketing and sports science, is also actively explored (Klink 2000;
16 Klink & Wu 2014; Shinohara et al. 2016). It is probably fair to say that the number of studies
17 on sound symbolism—and related topics, including iconicity and ideophones—is exponentially
18 growing in the last few decades (see Nielsen & Dingemanse 2020).

19 Against the background of this theoretical development, Kawahara et al. (2018) initiated a re-
20 search program, now dubbed “Pokémonastics” (Shih et al. 2019), in which researchers explore the
21 nature of sound symbolic connections in natural languages by studying Pokémon names. The first
22 Pokémonastics study explored sound symbolic patterns that hold among the existing Pokémon
23 names in Japanese, and found that several parameters—such as weight, height and strength—
24 correlate with the number of voiced obstruents contained in the names as well as the name lengths
25 quantified in terms of mora counts (Kawahara et al. 2018). Subsequent studies have extended the
26 scope of target languages, by analyzing existing Pokémon names and by conducting experiments
27 using nonce names. Building upon these previous studies, this paper reports a new experimental
28 study of Pokémonastics with Russian speakers, exploring what kinds of sound symbolic connec-
29 tions Russian speakers use to express the evolution status in the Pokémon universe.

30 While there are several research advantages of using Pokémon names to study sound sym-
31 bolic connections in natural languages, as recently summarized by Kawahara & Breiss (2021) and
32 Kawahara et al. (2021), one distinct advantage is that the set of denotations is fixed across lan-
33 guages in the Pokémon universe. Such is usually not the case in natural languages, as languages
34 can differ in the set of references to which they assign a name. Japanese, for example, needs to
35 distinguish “young sister” from “old sister,” and there is no age-neutral term which corresponds
36 to the English word “sister.” While this sort of complication is not insurmountable as we study
37 sound symbolism using existing words (e.g. basic vocabulary) (Blasi et al. 2016; Johansson et al.
38 2020; Wichmann et al. 2010), using Pokémon names allows us to circumvent this complication
39 altogether, because no matter which language we analyze, the set of denotations is fixed. In short,
40 allowing us to explore cross-linguistic similarities and differences in a systematic fashion is one
41 distinct forte of the Pokémonastics research paradigm (Shih et al. 2019). To this end, Shih et al.

(2019) analyzed sound symbolic patterns in the Pokémon lexicons of several languages, including Cantonese, English, Japanese, Korean, Mandarin and Russian. Several experimental studies using nonce words have also deployed the Pokémon universe to explore sound symbolic knowledge that speakers of various languages have, including English (Kawahara & Breiss 2021; Kawahara & Moore 2021), Japanese (Kawahara & Kumagai 2019; Kumagai & Kawahara 2019) and Brazilian Portuguese (Godoy et al. 2020).

These experimental studies have largely focused on the notion of evolution¹ —in the Pokémon world, characters can evolve into different creatures, and when they do, they tend to become larger, heavier and stronger, and they are assigned a new name. These previous studies have identified intriguing cross-linguistic similarities in terms of the sound symbolic associations that speakers of different languages make to express the evolution status. For example, speakers of all three languages find names with voiced obstruents to be more suitable for post-evolution characters than names with voiceless obstruents. This sound symbolic pattern may have its roots in the sound symbolic connection between voiced obstruents and largeness, which may have a phonetic basis. Since voiced obstruents are characterized by low frequency energy (Kingston & Diehl 1994; Stevens & Blumstein 1981), the physics tells us that they are produced by a large sound source (Ohala 1994).² Another cross-linguistically robust finding is that speakers of all these languages find names with [a] to be more suitable for post-evolution characters than those names with [i]. This finding reflects a very well-known sound symbolic effect in which [a] is judged to be larger than [i] (Sapir 1929; Westbury et al. 2018), presumably because [a] has low fundamental frequency and second formant frequency (Whalen & Levitt 1995). Again, the physics tells us that [a] is produced by a large sound source, whereas [i] is produced by a small sound source (Ohala 1994).

Our aim is to build upon these previous studies and expand the scope of Pokémonastics by studying what kinds of sound symbolic connections Russian speakers may make. Recall that allowing us to perform a controlled cross-linguistic comparison is one research advantage of Pokémonastics (Shih et al. 2019); therefore, we were particularly interested in testing the sound symbolic patterns that were identified in the previous experimental Pokémonastics studies which were conducted with speakers of other languages (Godoy et al. 2020; Kawahara & Breiss 2021; Kawahara & Moore 2021; Kawahara & Kumagai 2019; Kumagai & Kawahara 2019).

1.1 The current experiment

To this end, the current experiment tested four specific hypotheses:

¹Another semantic dimension that is actively explored via experimentation is the Pokémon types. See Kawahara et al. (2021) and references cited therein.

²Alternatively, voiced obstruents may be associated with large images because of the expansion of the oral cavity that is necessitated during their articulation (Ohala 1983).

(1) The hypotheses tested

- a. Names that contain coronal and dorsal consonants are judged to be more appropriate for post-evolution characters than names with labial consonants.
- b. Names that contain voiced obstruents are judged to be more appropriate for post-evolution characters than names with voiceless obstruents.
- c. Names that contain [a] are judged to be more appropriate for post-evolution characters than names with [i].
- d. Names that contain [r] (a trill) are judged to be more appropriate for post-evolution characters than names with [l].

The first hypothesis was motivated by the sound symbolic patterns that seem to hold in the existing Pokémon names in Russian. Shih et al. (2019) show that those characters which contain more coronal and dorsal consonants tend to be heavier. Since evolved characters are generally heavier in the Pokemon universe, we recast our hypothesis in terms of the evolution status, so that our results can be compared with the previous Pokémonastics experiments, which mainly explored sound symbolic patterns that are associated with the evolution status. The second and third hypotheses were motivated by the previous experimental studies of Pokémonastics reviewed above. Recall that speakers of three different languages—English, Japanese, and Brazilian Portuguese—tended to associate names with voiced obstruents and those with [a] with post-evolution characters (Godoy et al. 2020; Kawahara & Breiss 2021; Kawahara & Kumagai 2019; Kumagai & Kawahara 2019). The final hypothesis was motivated by an intuition that our native informant shared with us. She found that names with a trill may be more suited to express evolvedness in Russian than names with a lateral. Since this contrast was never tested in the previous Pokémonastics studies, we included this condition in the current experiment, as it may potentially expand the inventory of sound symbolic patterns that can be studied in the Pokémonastics research.

2 Methods

Following the spirit of open science initiative in linguistics and psychology (Chambers 2017; Winter 2019), the data, analysis code, and Bayesian posterior samples are made available at https://osf.io/j8brn/?view_only=127b8a9fe44e457c8478fc76bd4394b3. The R markdown files also contain visual illustrations of posterior distributions and conditional effects as well as posterior predictive checks.

2.1 Stimuli

The current experiment consisted of two sets of stimuli. In the first set, three factors (consonant place of articulation, consonant voicing, and vowel quality) were fully crossed. Place of articulation consisted of a contrast between labials, coronals, and dorsals. Consonants were either voiced or voiceless. Vowels were varied between [a] and [i]. This fully crossed design allowed us to address the additional question of whether or not sound symbolism functions in a cumulative way (Kawahara & Breiss 2021; Klink & Wu 2014), i.e. whether each factor can contribute to sound symbolic judgments independently of other factors, or whether one particular salient factor determines the sound symbolic value of the whole names.³

In the second set of stimuli, we tested the difference between a trill and a lateral, which was again crossed with the [a] vs. [i] difference. In both sets of the stimuli, the target sounds always appeared in the first syllable of the stimuli, because word-initial syllables generally convey strong sound symbolic effects (Adelman et al. 2018; Kawahara et al. 2008; Klink & Wu 2014). Three items were prepared within each cell by making use of different pseudo-suffixes. All the stimuli were disyllabic nonce words in Russian. The actual lists of stimuli appear in Tables 1 and 2. There were 36 items in total for the first set and 12 items for the second set.

Table 1: The first stimulus set.

[a]	<i>labial</i>	<i>coronal</i>	<i>dorsal</i>
<i>vls</i>	pamal	tamal	kamal
	pasma	tasma	kasma
	palfa	talfa	kalfa
<i>vcd</i>	bamal	damal	gamal
	basma	dasma	gasma
	balfa	dalfa	galfa
[i]	<i>labial</i>	<i>coronal</i>	<i>dorsal</i>
<i>vls</i>	pimal	timal	kimal
	pisma	tisma	kisma
	pilfa	tilfa	kilfa
<i>vcd</i>	bimal	dimal	gimal
	bisma	disma	gisma
	bilfa	dilfa	gilfa

³These two possibilities roughly correspond to Harmonic Grammar with weighted constraints (Legendre et al. 1990) and Optimality Theory with ranked constraints (Prince & Smolensky 1993/2004), two competing analytical frameworks in contemporary phonological studies. They also correspond to a regression-based decision making strategy and a fast-and-frugal heuristic decision making strategy (Gigerenzer & Gaissmaier 2011). See Kawahara & Breiss (2021) for extended discussion on this point.

Table 2: The second stimulus set.

	<i>trill</i>	<i>lateral</i>
[a]	rasim	lasim
	rahen	lahen
	rafos	lafos
[i]	risim	lisim
	rihen	lihen
	rifos	lifos

2.2 Procedure

The experiment was conducted online using SurveyMonkey. Participants agreed to participate in the experiment by reading through a consent form. The participants were told that some Pokémon characters undergo evolution, and when they do so, they tend to get bigger, heavier, and stronger. One stimulus name was presented per each trial and the participants were asked to choose whether the name was more appropriate for a pre-evolution name or a post-evolution name. The instructions as well as the stimuli were presented in the Cyrillic script. The order of 48 stimuli was randomized per participant. After the experiment, the participants were asked whether they were native speakers of Russian, whether they had studied sound symbolism before, and whether they had participated in any Pokémonastics experiments before.

2.3 Participants

The responses were collected using a snow-balling sampling method via word of mouth. A total of 849 speakers completed the online experiment without any monetary compensation.⁴ Data were excluded if they reported that they were a non-native speaker of Russian or if they knew sound symbolism or participated in a Pokémonastics experiment before ($N=122$). The data from the remaining 725 speakers entered in the subsequent statistical analyses. Among them, 421 of them were male, 294 of them were female and 10 did not report their gender. The age breakdown was as follows: 140 (18-19 yrs old), 357 (20-29 yrs old), 176 (30-39 yrs old), 36 (40-49 yrs old), 13 (50-59 yrs old) and 3 (60+ yrs old). Since there were no apparent effects of age or gender, and since we had no prior reasons to expect that these factors would impact the results (see Kawahara et al. 2020), we will not consider these factors in further detail.

⁴This happens to illustrate another general advantage of Pokémonastic experiments—we tend to be able to get responses from many participants even without monetary compensation, because people get curious what the experiment about Pokémon names is like.

2.4 Analyses

We fit a Bayesian mixed effects logistic regression model to analyze each data set. In both models, we included all the fixed factors as well as their interactions. Bayesian analyses have several advantages over more traditional frequentist analyses, which we do not review in detail here (see e.g. Franke & Roettger 2019 and Kruschke & Liddell 2018). It suffices to note that Bayesian analyses take prior information and the data under consideration to yield posterior distributions for each parameter that we are interested in. A straightforward heuristic to interpret the results of Bayesian regression analyses is to examine the 95% Credible Interval (CrI) of each coefficient—if this interval does not include 0, we can conclude that the effect is meaningful or credible. On the other hand, when a 95% CrI contains 0, we conclude that the effect is not very meaningful. Moreover, in Bayesian logistic regression models, if a CrI is fully contained in $[-0.18, 0.18]$, we can take that result to support a null effect (Kruschke & Liddell 2018; Makowski et al. 2019).⁵ We should note, however, that in Bayesian analyses, we can go beyond the usual “significant vs. non-significant” dichotomy, and directly explore the degrees of (un-)certainty of the estimates we are interested in.

The analyses were implemented using R (R Development Core Team 1993–) and the `brms` package (Bürkner 2017). In the current analyses, the random structure included a free-varying intercept and slope for participants associated with all the fixed factors and their interactions.⁶ The binary-coded dependent variable was whether the response was pre-evolution character or post-evolution character. The weakly informative prior, the default for `brms`, was used. Four chains were run with 4,000 iterations for each chain and 1,000 warmups. All the $\hat{R}$ -hat values associated with the fixed effects were 1.00 and there were no divergent transitions, indicating that the chains mixed successfully. See the R markdown files in the osf repository for complete details.

3 Results

Figure 1 shows the by-participant averages of each condition for the first set of stimuli. The y-axis represents the post-evolution responses. First, comparing the three columns, we observe that labial consonants generally induced fewer post-evolution responses than coronal and dorsal consonants (overall averages: 0.35 vs. 0.39 vs. 0.40). Second, comparing the top panels and bottom panels, we observe that voiced stops induced more post-evolution responses than voiceless stops (0.43

⁵This is a region that can be considered to be practically equivalent to 0, i.e. Region Of Practical Equivalence (a.k.a. ROPE) of $\beta=0$. In frequentist analyses, we are unable to “conclude the null” (Gallistel 2009). In Bayesian framework, on the other hand, we can distinguish cases that support the null effect from those cases that are not conclusive.

⁶The models did not include a free-varying random parameter for items, because item was a three-level factor and it does not seem appropriate to include a three-level random factor (see e.g. Snijders & Bosker 2011).

169 vs. 0.33). Finally, within each panel, we can see that names with [a] induced more post-evolution
 170 responses than those with [i] (0.44 vs. 0.32).

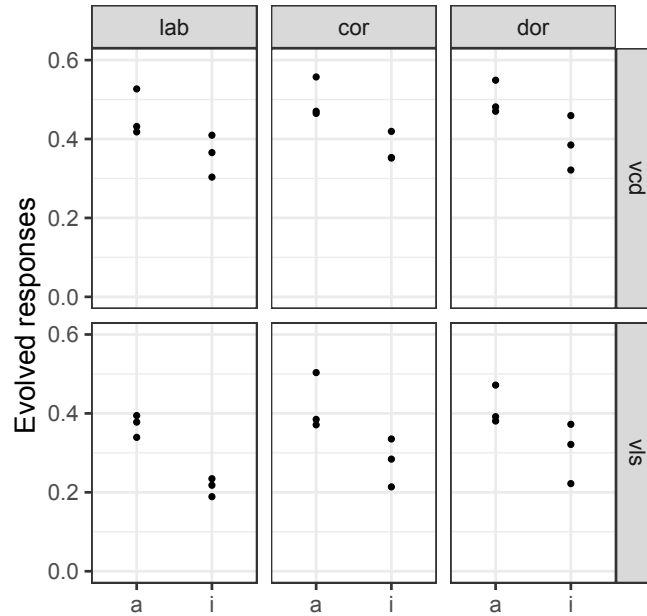


Figure 1: The results of the first set of stimuli.

171 The model summary of the regression analysis for this data set appears in Table 3. The baseline
 172 condition is voiced, [a] and coronal. First, the coefficient for the vls vs. vcd comparison is negative
 173 and its 95% CrI does not include 0, indicating that voiceless consonants meaningfully lowered
 174 post-evolution responses with respect to voiced consonants. Second, [i] lowered post-evolution
 175 responses with respect to [a], and again its 95% CrI does not include 0. Third, there does not seem
 176 to be a substantial difference between dorsal consonants and coronal consonants. Since the 95%
 177 CrI of this coefficient estimate is fully contained in [-0.18, 0.18], this result supports the thesis
 178 that coronals and dorsals are practically equivalent (Kruschke & Liddell 2018; Makowski et al.
 179 2019). Finally, there was a meaningful difference between labials and coronals, in such a way that
 180 labials lowered post-evolution responses with respect to coronals. In short, we observe that voiced
 181 obstruents and [a] tend to be associated with post-evolution characters, whereas labial consonants
 182 tend to be associated with pre-evolution characters.

183 All the interaction terms, except for the three-way interaction term between vls: [i]: lab, showed
 184 95% CrIs which include 0. The last interaction term indicates that the interaction between con-
 185 sonant voicing and the vowel quality is slightly less pronounced for labials. Setting aside this
 186 complicated interaction effect, these results show that the effects of the three factors—place of ar-
 187 ticulation, voicing and vowel quality—were generally additive; i.e. they more or less contributed to

Table 3: The model summary (the first set of stimuli).

	β	error	95% CrI
intercept	-0.06	0.07	[-0.19, 0.08]
vls (vs. vcd)	-0.39	0.07	[-0.53, -0.25]
[i] (vs. [a])	-0.60	0.08	[-0.76, -0.45]
dor (vs. cor)	0.02	0.07	[-0.11, 0.16]
lab (vs. cor)	-0.20	0.07	[-0.34, -0.06]
vls: [i]	-0.15	0.10	[-0.35, 0.04]
vls: dor	-0.04	0.10	[-0.23, 0.16]
vls: lab	-0.07	0.10	[-0.27, 0.12]
[i]: dor	0.06	0.10	[-0.12, 0.26]
[i]: lab	0.11	0.10	[-0.08, 0.31]
vls: [i]: dor	0.15	0.14	[-0.13, 0.42]
vls: [i]: lab	-0.32	0.15	[-0.61, -0.02]

188 post-evolution responses independently of one another, although we should bear in mind that since
189 the CrIs for the interaction terms are not fully contained in [-0.18, 0.18], the current results do not
190 support the null effect for these interaction terms. See Kawahara & Breiss (2021) who deployed a
191 similar experimental design and obtained similar results with English speakers.

192 Figure 2 shows the results of the second set of stimuli. As was the case in Figure 1, the dif-
193 ference between [a] and [i] was apparent in that the former induced more post-evolution responses
194 (0.57 vs. 0.49). The effect of the difference between a trill and a lateral was in the expected direc-
195 tion, but not very large in size (0.54 vs. 0.51).

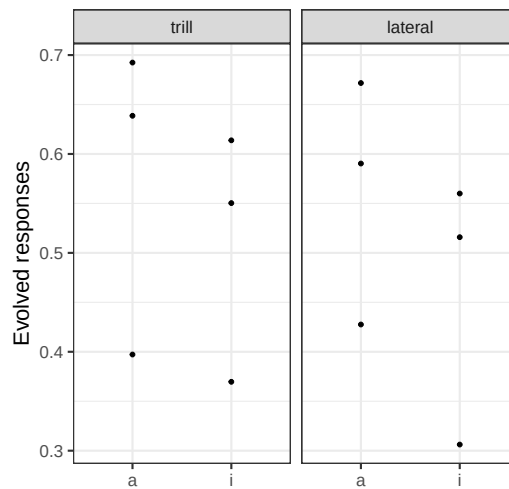


Figure 2: The results of the second set of stimuli.

The model summary of the regression analysis for this date set appears in Table 4. The difference between the two liquids was not credible; neither was its interaction with the vowel effect. On the other hand, [i] credibly lowered post-evolution responses compared to [a].

Table 4: The model summary (the second set of stimuli).

	β	error	95% CrI
intercept	0.29	0.06	[0.18, 0.39]
trill (vs. lateral)	0.06	0.07	[-0.06, 0.19]
[i] (vs. [a])	-0.47	0.07	[-0.60, -0.34]
[i]: trill	0.17	0.09	[-0.01, 0.35]

Since the 95% CrI for the interaction term is skewed toward positive (i.e. its lower bound is -0.01), we further explored how many posterior samples of this interaction term were positive, and found that 96.6% of them are positive. This implies that the effect of the trill vs. lateral difference is more pronounced in the [i]-condition than in the [a]-condition (overall averages: 0.58 vs. 0.56 in the [a]-condition and 0.51 vs. 0.46 in the [i]-condition). Since the slope estimate for the effect of trills in the [i] condition can be expressed as the sum of the effect of trills and the interaction term (see Franke & Roettger 2019 and Winter 2019: 143-143 for nice illustrations), we calculated the proportion of posterior probabilities of the sum of these two coefficients being positive. This analysis shows that 99.9% of them are positive, suggesting that there was an effect of trills in the context of [i]. In sum, trills were more likely to be associated with post-evolution characters than laterals, but only before [i]. We do not have a good explanation for why this complicated pattern holds.

4 Discussion

To summarize the main findings of the current experiment, Russian speakers found (i) names with [a] to be more suitable for post-evolution characters than names with [i], (ii) names with voiced obstruents to be more suitable for post-evolution characters than those with voiceless obstruents, (iii) names with labial consonants to be less suitable for post-evolution characters than those with coronal and dorsal consonants, and (iv) the difference between the two types of liquids was limited when they are followed by [i].

We find the first two findings to be very encouraging. Recall that these connections have been found in previous experimental Pokémon studies, targeting speakers of different languages, including English (Kawahara & Breiss 2021), Japanese (Kawahara & Kumagai 2019; Kumagai & Kawahara 2019) and Brazilian Portuguese (Godoy et al. 2020). The current results are all the

more intriguing because these patterns do not seem to hold in the names used for Russian Pokémon characters (Shih et al. 2019). It is thus likely that these sound symbolic patterns arose in the current experiment because of knowledge of sound symbolism that Russian speakers have, which may be phonetically grounded—voiced obstruents are judged to be large because of their low frequency energy; [a] is likewise judged to be large because it has low fundamental frequency as well as low second formant frequency (Ohala 1994). We note, however, that the fact that [i] induced fewer post-evolution responses may at least in part be affected by certain aspects of Russian phonology: [i] causes palatalization of the preceding consonants (Padgett 2001) and palatalized consonant can denote diminutive meanings (Alderete & Kochetov 2017); e.g. [lʲaɫʲa] is the child word meaning “baby” or “doll” (Shih et al. 2019). A follow-up experiment can tease apart the effects of [i] and those of palatalized consonants by comparing three conditions: [Cʲi], [Cʲa] and [Ca].

The fact that labial consonants were less likely to be associated with post-evolution characters, compared to coronal and dorsal consonants, may have arisen as a result of inferences from the pattern in the existing Pokémon names in Russian (Shih et al. 2019). An alternative possibility, which is not incompatible with this first explanation, is that labial consonants may inherently invoke small and cute images (Kumagai & Kawahara 2020), because labial consonants are those that are observed in babbling and early speech i.e. those sounds that are prototypically produced by babies (Jakobson 1941; MacNeilage et al. 1997). This sound symbolic effect has been observed with Japanese speakers (Kumagai & Kawahara 2020) as well as with English speakers (Kawahara & Moore 2021), another case of cross-linguistic similarity that we find very intriguing.

It is not immediately clear why the sound symbolic effect of trills was limited to a particular vocalic environment. It is possible that trills in word-initial positions may not realize as prototypical trills, possibly for an aerodynamic reason (Solé 2002), so that its effect is small and can only be detected in a certain vocalic environment. It is not still clear, however, why the [i]-context specifically modulated the effects of trills. The sound symbolic values of trills, or lack thereof, should be tested in follow-up experiments with stimuli containing trills in different positions as well as other vocalic environments.⁷

Finally, the three sound symbolic patterns that we identified in the first set of the stimuli (place of articulation, voicing and vowel) seem to generally function in a cumulative fashion—each factor contributed to the post-evolution responses in an additive way, instead of each particular segment determining the sound symbolic value of the whole names. This finding adds to some recent findings that sound symbolic patterns generally function in a cumulative way (Kawahara 2020; Kawahara & Breiss 2021; Klink & Wu 2014).

⁷Another topic that we would like to explore in future experimentation is the effect of “the iconicity of quantity” (Haiman 1984), in which longer names are more likely to be associated with larger post-evolution characters. This effect is observed robustly across languages in previous Pokémon studies (Godoy et al. 2020; Kawahara & Breiss 2021; Kawahara & Kumagai 2019).

Author contributions

The first author conceived, designed and executed the experiment with the help of a native speaker consultant. The second author discussed the overall experimental design, statistically analyzed the data and wrote up the first draft of the manuscript. The first author extensively revised the manuscript.

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